# MidTerm

February 16, 2026

## Problem 1

Consider a softmax model over categories, with a total of $C$ possible categories. Assume that the probability of the observed category $y \in \{1, \ldots, C\}$ is

$$p(Y = y|F = f) = \frac{\exp(w_y^T f)}{\sum_{c=1}^{C} \exp(w_c^T f)}\,, \tag{1}$$

where $f \in \mathbb{R}^{d'}$, and each $w_c \in \mathbb{R}^{d'}$, for $c \in \{1, \ldots, C\}$. Assume that the set of vectors $\{w_c\}_{c=1,C}$ is known and fixed.

(a) Assume that we are given the observed category $y \in \{1, \ldots, C\}$, and our objective is to infer the associated latent vector $f \in \mathbb{R}^{d'}$.

**Show** that the *cross entropy* (negative log-likelihood) loss $\ell(f)$ for this observation $Y = y$, with $F = f$ as the unknown parameters we desire, may be expressed as

$$\ell(f) = -w_y^T f + \log \sum_{c=1}^{C} \exp(w_c^T f)\,.$$

(b) Assume that we *initialize* $f = f_0$, and that we seek to update it with one step of gradient descent.

**Show** that one step of gradient descent yields the update

$$f_1 = f_0 + \alpha \left[ w_y - \mathbb{E}(w)_{|f_0} \right]\,, \tag{2}$$

where $w_y$ is the vector within the softmax probability model corresponding to observed category $y \in \{1, \ldots, C\}$,

$$\mathbb{E}(w)_{|f_0} = \sum_{j=1}^{C} w_j \left[ \frac{\exp(w_j^T f_0)}{\sum_{c=1}^{C} \exp(w_c^T f_0)} \right]\,, \tag{3}$$

and $\alpha > 0$ is the learning rate.

(c) Consider a model as reflected in Figure 1, in which a feedforward network $g_\theta(f_0, w_y)$ operates on the concatenated input vector $(f_0, w_y)$, where $f_0$ is the initialized vector for $f$ and the vector $w_y$ is associated with observed category $y \in \{1, \ldots, C\}$. The function is designed such that the output of $g_\theta(f_0, w_y)$ can be expressed as $(f_{out}, 0_{d'})$, where $0_{d'}$ is a $d'$-dimensional vector of all zeros, and $f_{out}$ is a computed output vector of dimension $d'$. With the skip connection we manifest

$$(f_0, w_y) \to (f_0 + f_{out}, w_y)\,.$$

The parameters $\theta$ are connected to the neural network represented by $g_\theta(f_0, w_y)$. The vector $f = f_0 + f_{out}$ is input to the softmax function to predict the output probability over categories is

$$p(Y = y|F = f) = \frac{\exp(w_y^T f)}{\sum_{c=1}^{C} \exp(w_c^T f)}\,.$$

*Our goal is to learn the model parameters* $\theta$ such that the ultimate $f = f_0 + f_{out}$ that feeds into the softmax at the output yields high probability for the observed category $y \in \{1, \ldots, C\}$. With $f = f_0 + f_{out}$, note

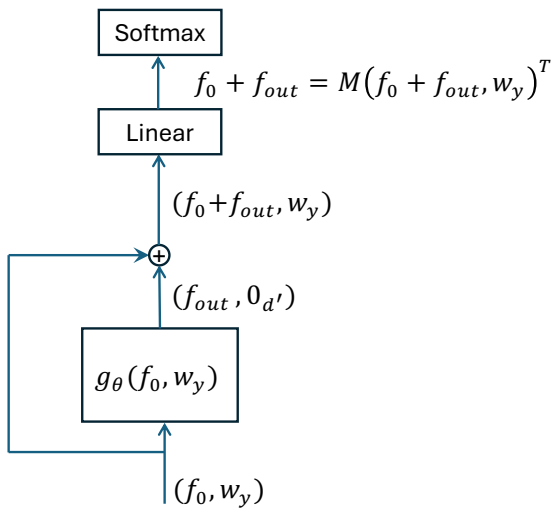


Figure 1: Skip connection model with an isolated feedforward element $g_\theta(f_0, w_y)$. The input to the network $g_\theta(f_0, w_y)$ is the *initialized* vector $f_0 \in \mathbb{R}^{d'}$, and the vector $w_y \in \mathbb{R}^{d'}$ associated with the observed category $y \in \{1, \ldots, C\}$ (one of the vectors $\{w_c\}_{c=1,C}$ within the softmax model, for the observed category $y \in \{1, \ldots, C\}$). The *output* of $g_\theta(f_0, w_c)$ is a $d'$-dimensional vector $f_{out}$, concatenated with appended $d'$ zeros, yielding the cumulative output $(f_{out}, 0_{d'})$. The output after adding the skip connection is $(f_{out} + f_0, w_y)$. This is then multiplied by the matrix $M \in \mathbb{R}^{d' \times 2d'}$, with $M = \left[I_{d' \times d'} \; 0_{d' \times d'}\right]$, to yield the final output $f = f_{out} + f_0$ which feeds into the softmax function in (1).

that $f_{out}$ is a function of the parameters $\theta$ in $g_\theta(f_0, w_y)$. Hence, imposing properties on $f_{out}$ corresponds to imposing conditions on the model parameters $\theta$.

**Show** that the negative log-likelihood for the model parameters is, to first order,

$$\ell(\theta) = -\log p(Y = y | f_0 + f_{out}) \approx -\log p(Y = y | f_0) - f_{out} \cdot [\nabla_f \log p(Y = y | f)]_{f=f_0} \,.$$

where $f_{out} \cdot [\nabla_f \log p(Y = y|f)]_{f=f_0}$ is the inner product of $f_{out}$ and the vector defined by $[\nabla_f \log p(Y = y|f)]_{f=f_0}$. Recall that the first two terms in a Taylor expansion of $f(x_0 + \Delta)$ are $f(x_0 + \Delta) \approx f(x_0) + \Delta[\frac{df(x)}{dx}]_{|x=x_0}$, where $[\frac{df(x)}{dx}]_{|x=x_0}$ means that the derivative is evaluated at $x = x_0$.

(d) Based on the above results, to minimize the negative log likelihood for learning $\theta$, show that ideally the vector output $f_{out}$ from the feedforward model $g_\theta(f_0, w_y)$ should be in the direction of the vector $[\nabla_f \log p(Y = y|f)]_{f=f_0}$. In other words, to minimize the loss, the inner product between $f_{out}$ and $[\nabla_f \log p(Y = y|f)]_{f=f_0}$ should be as large as possible, implying that ideally

$$f_{out} = \alpha[\nabla_f \log p(Y = y|f)]_{f=f_0} = \alpha[w_y - \mathbb{E}(w)_{|f=f_0}] \,,$$

where $\alpha > 0$.

**Explain/justify** why we expect this. Further, **show** that the output of the feedforward element plus skip connection is ideally (to first order)

$$f_0 + f_{out} \approx f_0 + \alpha[w_y - \mathbb{E}(w)_{|f=f_0}] \,,$$

*i.e.*, one step of gradient descent for the vector $f$, starting at the initial $f_0$. Hence, ideally, the feedforward element $g_\theta(f_0, w_y)$ should compute $\alpha[w_y - \mathbb{E}(w)_{|f=f_0}]$, to first order.

## Problem 2

Consider contextual data $\{(x_i, y_i)\}_{i=1,N}$, where $x_i \in \mathbb{R}^d$ and $y_i \in \{1, \ldots, C\}$. We are also given a query $x_{N+1} \in \mathbb{R}^d$, for which we wish to infer the associated category $y_{N+1} \in \{1, \ldots, C\}$. Assume that there is an underlying function $f(x) \in \mathbb{R}^{d'}$, and the probability of category $y \in \{1, \ldots, C\}$ is

$$p(Y = y|f(x)) = \frac{\exp(w_y^T f(x))}{\sum_{c=1}^{C} \exp(w_c^T f(x))}\,, \tag{4}$$

where $\{w_c\}_{c=1,C}$ are fixed and known.

Assume that the underlying function $f(x) = A\phi(x)$, where $\phi(x) \in \mathbb{R}^n$ is a fixed nonlinear transformation of $x$ to an $n$-dimensional feature space, and $A \in \mathbb{R}^{d' \times n}$ is an unknown, context-dependent *matrix*. If $a_m \in \mathbb{R}^n$ is row $m$ of $A$, then the $m$th component of vector $f(x)$, is $f_m(x) = a_m \phi(x)$. Additionally, define the kernel $\kappa(x_i, x_j) = \phi(x_i)^T \phi(x_j)$.

(a) Using the contextual data $\{(x_i, y_i)\}_{i=1,N}$ and the model for $p(Y = y|f(x))$.
**Derive** the negative log-likelihood for the unknown, context-dependent matrix $A$.

(b) Recalling that $a_m \in \mathbb{R}^n$ represents the $m$th row of $A$,
**Show** that one gradient-descent update equation for the weights $a_m$ satisfies

$$a_m^{(1)} = a_m^{(0)} + \alpha \sum_{i=1}^{N} [w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}}]\phi(x_i)\,, \tag{5}$$

where $w_{y_i}$ corresponds to the vector within the softmax of (4) for category $y_i \in \{1, \ldots, C\}$, and $w_{y_i}(m)$ represents component $m$ of $w_{y_i}$. The vector $a_m^{(0)}$ is the *initial* setting of the $m$th row of $A$, and $f^{(0)}(x) = A^{(0)}\phi(x)$, where $A^{(0)}$ represents the initial underlying matrix (prior to implementing gradient descent). The parameter $\alpha > 0$ is the learning rate. Further,

$$\mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} = \sum_{c=1}^{C} w_c(m) \frac{\exp(w_c^T f^{(0)}(x_i))}{\sum_{c'=1}^{C} \exp(w_{c'}^T f^{(0)}(x_i))}\,.$$

<u>Hint</u>: To calculate the gradient of the loss function $\nabla_{a_m} \ell(a_m)$, consider the chain rule $\nabla_{a_m} \ell(a_m) = [\nabla_{a_m} f_m(x)] \frac{\partial \ell[f(x)]}{\partial f_m(x)}$.

(c) Using (5),
**Show** that the underlying function updates with GD steps as

$$f^{(1)}(x_j) = f^{(0)}(x_j) + \alpha \sum_{i=1}^{N} [w_{y_i} - \mathbb{E}[w]_{|_{f^{(0)}(x_i)}}]\kappa(x_i, x_j)\,.$$

(d) If the latent function is initialized $f^{(0)}(x) = 0_{d'}$ for all $x$,
**Show** that for all $x_i$

$$\mathbb{E}[w]_{|_{f^{(0)}(x_i)}} = \frac{1}{C} \sum_{c=1}^{C} w_c\,.$$

# Solutions

## Problem 1

(a) We are asked to show that the cross-entropy for the observation $Y = y$ can be expressed as

$$\ell(f) = -w_y^T f + \log \sum_{c=1}^{C} \exp(w_c^T f).$$

Starting with the probability definition:

$$p(Y = y|F = f) = \frac{\exp(w_y^T f)}{\sum_{c=1}^{C} \exp(w_c^T f)}.$$

The negative log-likelihood (cross-entropy) for the observed category $y$ is defined as:

$$\ell(f) = -\log p(Y = y|F = f).$$

Substituting $p(Y = y|F = f)$:

$$\ell(f) = -\log \left( \frac{\exp(w_y^T f)}{\sum_{c=1}^{C} \exp(w_c^T f)} \right).$$

Expanding the logarithm:

$$\ell(f) = -w_y^T f + \log \left( \sum_{c=1}^{C} \exp(w_c^T f) \right).$$

Hence, the desired form is shown.

(b) We are tasked with showing that one step of gradient descent yields the update:

$$f_1 = f_0 + \alpha[w_y - \mathbb{E}(w)_{|f_0}],$$

where

$$\mathbb{E}(w)_{|f_0} = \sum_{j=1}^{C} w_j \left[ \frac{\exp(w_j^T f_0)}{\sum_{c=1}^{C} \exp(w_c^T f_0)} \right].$$

The gradient of the loss $\ell(f)$ is:

$$\nabla_f \ell(f) = -w_y + \sum_{j=1}^{C} w_j \left[ \frac{\exp(w_j^T f)}{\sum_{c=1}^{C} \exp(w_c^T f)} \right].$$

Evaluating this at $f = f_0$:

$$\nabla_f \ell(f)\big|_{f=f_0} = -w_y + \mathbb{E}(w)_{|f_0}.$$

The gradient descent update rule is:

$$f_1 = f_0 - \alpha \nabla_f \ell(f)\big|_{f=f_0}.$$

Substituting the gradient:

$$f_1 = f_0 - \alpha(-w_y + \mathbb{E}(w)_{|f_0}).$$

Simplifying:

$$f_1 = f_0 + \alpha[w_y - \mathbb{E}(w)_{|f_0}].$$

(c) Given the skip connection model in Figure 1, the negative log-likelihood for the model parameters $\theta$ is:

$$\ell(\theta) = -\log p(Y = y | f_0 + f_{\text{out}}).$$

Using a Taylor expansion around $f = f_0$:

$$\log p(Y = y | f_0 + f_{\text{out}}) \approx \log p(Y = y | f_0) + f_{\text{out}} \cdot \nabla_f \log p(Y = y | f)\big|_{f=f_0}.$$

Thus:

$$\ell(\theta) \approx -\log p(Y = y | f_0) - f_{\text{out}} \cdot \nabla_f \log p(Y = y | f)\big|_{f=f_0}.$$

(d) To minimize $\ell(\theta)$, we need to maximize the inner product $f_{\text{out}} \cdot \nabla_f \log p(Y = y|f)\big|_{f=f_0}$. The gradient of the log-probability at $f = f_0$ is:

$$\nabla_f \log p(Y = y|f)\big|_{f=f_0} = w_y - \mathbb{E}(w)_{|f_0}.$$

Thus, to maximize the inner product, we ideally set:

$$f_{\text{out}} = \alpha[w_y - \mathbb{E}(w)_{|f_0}],$$

where $\alpha > 0$.

With the skip connection, the output becomes:

$$f_0 + f_{\text{out}} \approx f_0 + \alpha[w_y - \mathbb{E}(w)_{|f_0}],$$

which corresponds to one step of gradient descent.

## Problem 2

(a) The negative log-likelihood for the data $\{(x_i, y_i)\}_{i=1}^N$ is:

$$\ell(A) = -\sum_{i=1}^{N} \log p(Y = y_i | f(x_i)),$$

where $f(x) = A\phi(x)$ and:

$$p(Y = y | f(x)) = \frac{\exp(w_y^T f(x))}{\sum_{c=1}^{C} \exp(w_c^T f(x))}.$$

Substituting $f(x_i) = A\phi(x_i)$:

$$\ell(A) = -\sum_{i=1}^{N} \log \left( \frac{\exp(w_{y_i}^T A\phi(x_i))}{\sum_{c=1}^{C} \exp(w_c^T A\phi(x_i))} \right).$$

Expanding the logarithm:

$$\ell(A) = -\sum_{i=1}^{N} \left[ w_{y_i}^T A\phi(x_i) - \log \left( \sum_{c=1}^{C} \exp(w_c^T A\phi(x_i)) \right) \right].$$

(b) Using the chain rule:

$$\nabla_{a_m} \ell(a_m) = \sum_{i=1}^{N} \left[ \frac{\partial \ell(f(x_i))}{\partial f_m(x_i)} \cdot \nabla_{a_m} f_m(x_i) \right].$$

The gradient of $f_m(x_i) = a_m^T \phi(x_i)$ with respect to $a_m$ is $\phi(x_i)$.
Thus:

$$\nabla_{a_m} \ell(a_m) = \sum_{i=1}^{N} \left( \frac{\partial \ell(f(x_i))}{\partial f_m(x_i)} \right) \phi(x_i).$$

The partial derivative $\frac{\partial \ell(f(x_i))}{\partial f_m(x_i)}$ is:

$$\frac{\partial \ell(f(x_i))}{\partial f_m(x_i)} = w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f(x_i)}}.$$

Hence:

$$\nabla_{a_m} \ell(a_m) = \sum_{i=1}^{N} \left[ w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f(x_i)}} \right] \phi(x_i).$$

The gradient descent update is:

$$a_m^{(1)} = a_m^{(0)} + \alpha \sum_{i=1}^{N} \left[ w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} \right] \phi(x_i).$$

(c) We are tasked with deriving the updated function $f_m^{(1)}(x_j)$ after one step of gradient descent. Starting from the update rule for $a_m$:

$$a_m^{(1)} = a_m^{(0)} + \alpha \sum_{i=1}^{N} \left[ w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} \right] \phi(x_i).$$

The function $f_m(x_j)$ is defined as $f_m(x_j) = a_m^T \phi(x_j)$. Substituting the updated $a_m^{(1)}$:

$$f_m^{(1)}(x_j) = \left( a_m^{(0)} + \alpha \sum_{i=1}^{N} \left[ w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} \right] \phi(x_i) \right)^T \phi(x_j).$$

Expanding this expression:

$$f_m^{(1)}(x_j) = f_m^{(0)}(x_j) + \alpha \sum_{i=1}^{N} \left[ w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} \right] \phi(x_i)^T \phi(x_j).$$

Using the kernel definition $\kappa(x_i, x_j) = \phi(x_i)^T \phi(x_j)$, we have:

$$f_m^{(1)}(x_j) = f_m^{(0)}(x_j) + \alpha \sum_{i=1}^{N} \left[ w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} \right] \kappa(x_i, x_j).$$

(d) When $f^{(0)}(x) = 0$, the softmax probabilities for all classes are equal:

$$p(Y = c|f^{(0)}(x_i)) = \frac{\exp(0)}{\sum_{c'=1}^{C} \exp(0)} = \frac{1}{C}.$$

The expected value $\mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}}$ is given by:

$$\mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} = \sum_{c=1}^{C} w_c(m) \cdot p(Y = c|f^{(0)}(x_i)).$$

Substituting $p(Y = c|f^{(0)}(x_i)) = \frac{1}{C}$:

$$\mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} = \sum_{c=1}^{C} w_c(m) \cdot \frac{1}{C} = \frac{1}{C} \sum_{c=1}^{C} w_c(m).$$

Thus, the expected value simplifies to the average of the weights over all classes:

$$\mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} = \frac{1}{C} \sum_{c=1}^{C} w_c(m).$$